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Department of Electronics and Communication Engineering

19EAC384 – Computer Networks Lab (S6 EAC)

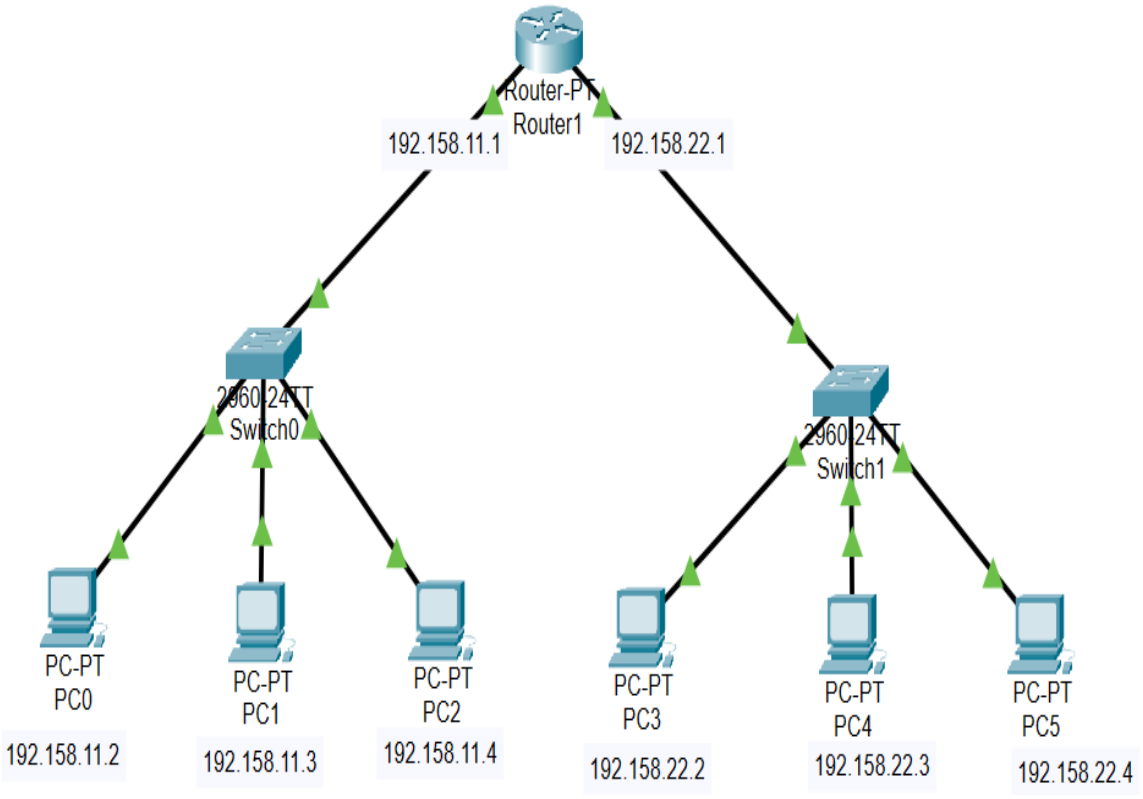
LABSHEET 4 - Assignment

BUILDING A WIDE AREA NETWORK (WAN) SET-UP

Aim:

- To build a WAN set up by configuring computers to form Local Area Networks (LAN) and establish communication between PCs in different LANs using Routers.
- To assign IP address dynamically to PCs using DHCP server.

Exercises-1: Build two simple LANs using 3 machines connected via star topology. Establish communication between two PCs belonging to different LANs using a single Router. In your document, highlight the steps adopted to set up the individual LANs and to configure the router and PCs to establish successful communication.



CLI Commands Listed:

```
R1(config-if)#
R1(config-if)#! Configure FastEthernet0/0 for Network 1
R1(config-if)#interface FastEthernet0/0
R1(config-if)#ip address 192.158.11.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
R1(config)#! Configure FastEthernet1/0 for Network 2
R1(config)#interface FastEthernet1/0
R1(config-if)#ip address 192.158.22.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
R1(config)#! Exit to privileged mode
R1(config)#end
R1#
R1#! Save the configuration
R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#! Verify the interfaces
R1#show ip interface brief
%SYS-5-CONFIG_I: Configured from console by console
```

Ping Commands:

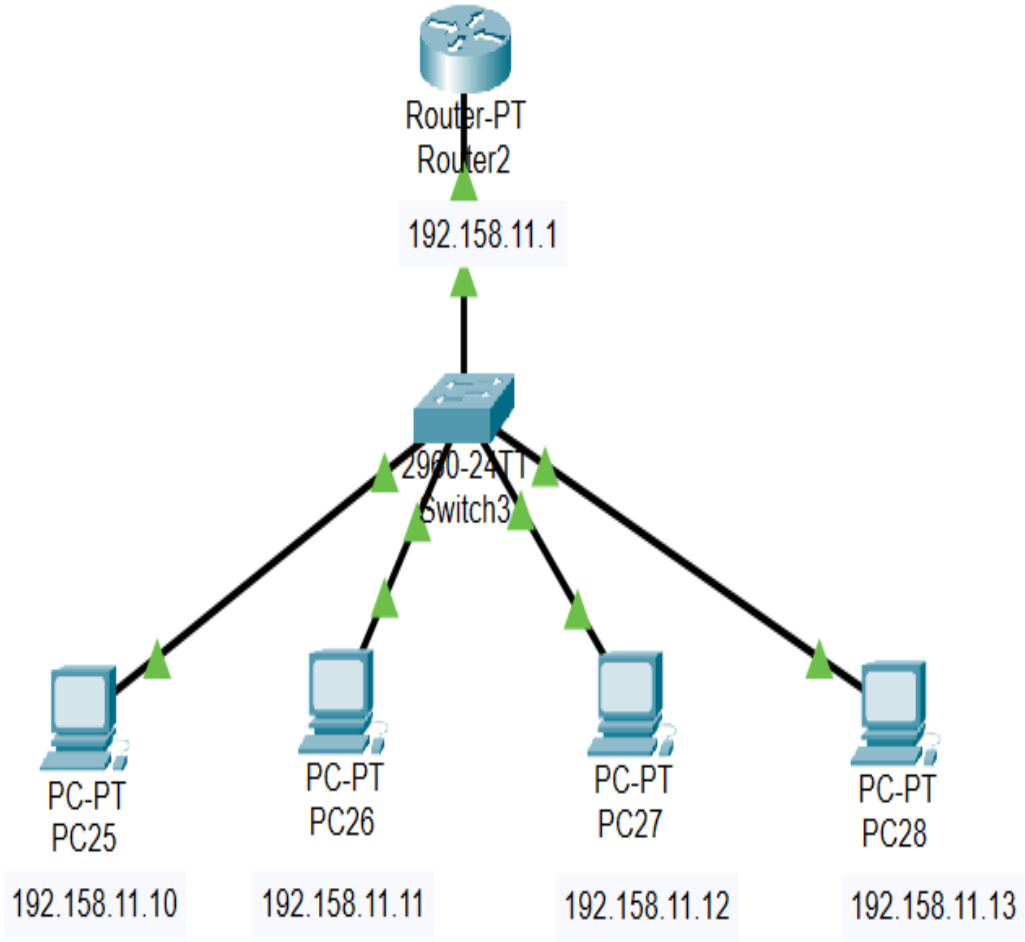
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.158.22.4

Pinging 192.158.22.4 with 32 bytes of data:

Reply from 192.158.22.4: bytes=32 time=6ms TTL=128
Reply from 192.158.22.4: bytes=32 time=3ms TTL=128
Reply from 192.158.22.4: bytes=32 time=4ms TTL=128
Reply from 192.158.22.4: bytes=32 time=4ms TTL=128

Ping statistics for 192.158.22.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 6ms, Average = 4ms
```

```
C:\>ping 192.158.11.3

Pinging 192.158.11.3 with 32 bytes of data:

Reply from 192.158.11.3: bytes=32 time<1ms TTL=128
Reply from 192.158.11.3: bytes=32 time<1ms TTL=128
Reply from 192.158.11.3: bytes=32 time<1ms TTL=128
Reply from 192.158.11.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.158.11.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Inference:

Configuring two LANs connected via a router using CLI commands offered a valuable learning experience in understanding IP configuration and routing. Initially, I faced some difficulties in enabling communication between PCs on separate networks. However, after thoroughly checking the configuration and verifying that the IP addresses, subnet masks, and default gateways were correctly assigned, the issue was resolved, and inter-network communication was successfully achieved. This practical activity emphasized the critical role of accurate network settings in facilitating communication between devices across different LANs through a router.

Exercises-2: To configure a DHCP server in Cisco Packet Tracer that assigns IP addresses dynamically to 3 PCs connected to a switch. In your document give your observations.



CLI Commands Listed:

```
Router(config)#
Router(config)#! Step 1: Exclude IPs from DHCP pool
Router(config)#ip dhcp excluded-address 192.158.11.1 192.158.11.9
Router(config)#
Router(config)#! Step 2: Create DHCP Pool
Router(config)#ip dhcp pool LAN_Pool
Router(dhcp-config)# network 192.158.11.0 255.255.255.0
Router(dhcp-config)# default-router 192.158.11.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#
Router(config)#! Step 3: Configure Router Interface (Facing the LAN)
Router(config)#interface FastEthernet0/0
Router(config-if)# ip address 192.158.11.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#end
Router#
Router#! Step 4: Save configuration
Router#copy running-config startup-config
Destination filename [startup-config]?
%SYS-5-CONFIG_I: Configured from console by console
```

Ping Commands:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.158.11.11

Pinging 192.158.11.11 with 32 bytes of data:

Reply from 192.158.11.11: bytes=32 time<1ms TTL=128
Reply from 192.158.11.11: bytes=32 time<1ms TTL=128
Reply from 192.158.11.11: bytes=32 time<1ms TTL=128
Reply from 192.158.11.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.158.11.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 192.158.11.12

Pinging 192.158.11.12 with 32 bytes of data:

Reply from 192.158.11.12 : bytes=32 time<1ms TTL=128
Reply from 192.158.11.12 : bytes=32 time<1ms TTL=128
Reply from 192.158.11.12 : bytes=32 time<1ms TTL=128
Reply from 192.158.11.12 : bytes=32 time<1ms TTL=128

Ping statistics for 192.158.11.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Inference:

Setting up DHCP using the command line on the router helped me understand how IP addresses are given out automatically in a network. At first, I had some trouble figuring out the right interface and how to create the DHCP pool correctly. But after trying different things and learning from my mistakes, I got it working. This experience made it easier to understand how IP addressing and router settings work, and it also made me feel more confident using the CLI in Cisco Packet Tracer.